

EA Before After Impact

Enterprise Architecture: Before vs After Impact

Purpose: Document the consequences and impact of implementing Enterprise Architecture in an organization

Overview

BEFORE EA vs AFTER EA	
<u>BEFORE EA (Chaos)</u>	<u>AFTER EA (Order)</u>
"Everyone does their own thing"	"We have standards and guidelines"
EA transforms CHAOS into ORDER!	

1. Technology Decisions

TECHNOLOGY DECISIONS	
<u>BEFORE EA:</u>	<u>AFTER EA:</u>
- Team A uses MySQL - Team B uses PostgreSQL - Team C uses MongoDB - Team D uses Oracle	- Standard: PostgreSQL for all - Approved technology catalog - Clear guidelines for selection - Exceptions need approval

RESULT: 4 different databases to maintain, integrate, secure	RESULT: Consistent stack Easier maintenance, lower cost
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Real Example

Metric	Before EA	After EA	Improvement
Message Brokers	7 different (Kafka, RabbitMQ, SQS, Redis Pub/Sub, ActiveMQ, ZeroMQ, custom)	2 (Kafka + SQS only)	70% reduction
Maintenance Effort	High	Low	40% reduction

2. Integration & Communication

INTEGRATION & COMMUNICATION

BEFORE EA:

- Point-to-point integrations
- No API standards
- Each team different format
- Spaghetti architecture

A → B

↓ ↓

C ↔ D → E

↑ ↓ ↓

F ↔ G ↔ H

(Spaghetti)

AFTER EA:

- Standard API patterns (REST/gRPC)
- API Gateway for all services
- Consistent contracts
- Clean, documented integrations

API Gateway

├── A

├── B

└── C

(Organized)

Real Example

file:///Users/ranadeeptujala/Desktop/workspace/togafea/EA_Before_After_Impact.html

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Metric	Before EA	After EA	Improvement
Integration time for new service	3-4 weeks	2-3 days	80% faster

3. Decision Making

DECISION MAKING	
BEFORE EA: • Decisions made in silos • No visibility across teams • Repeated mistakes • "Why did we choose this?" – Unknown	AFTER EA: • Architecture Review Board • Decisions documented (ADRs) • Learn from past decisions • Clear rationale recorded

Real Example

Scenario	Before EA	After EA
New developer asks "Why PostgreSQL?"	Nobody knows	Check ADR-001 → Full context, options considered, reasons

4. Project Delivery

PROJECT DELIVERY	
BEFORE EA: • Start from scratch every project • Long discovery phase • Inconsistent quality • Unpredictable timelines	AFTER EA: • Reuse building blocks (SBBs) • Reference architectures available • Standards ensure quality • Faster, predictable delivery

Real Example

Metric	Before EA	After EA	Improvement
New microservice development	6-8 weeks (research, design, build from scratch)	2-3 weeks (use templates, patterns, reusable components)	60% faster

5. Cost & Efficiency

COST & EFFICIENCY	
BEFORE EA: • Duplicate systems doing same thing • High licensing costs • Wasted infrastructure • Technical debt growing	AFTER EA: • Consolidation & rationalization • Reduced vendor sprawl • Right-sized resources • Managed technical debt

Real Example

Metric	Before EA	After EA	Improvement
Application portfolio	150+ applications (many doing similar things)	85 applications (consolidated, rationalized)	40% reduction
Annual savings	-	\$2M/year in licensing + maintenance	Significant

6. Risk & Security

RISK & SECURITY	
BEFORE EA:	AFTER EA:

• Inconsistent security practices • Unknown vulnerabilities • No visibility of risks • Compliance gaps	• Security principles for all • Standard security patterns • Risk assessment in architecture • Compliance by design
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Real Example

Metric	Before EA	After EA	Improvement
Security audit findings	45 critical issues	5 issues (all in legacy systems)	90% reduction
New systems	Inconsistent security	Secure by default (follows security architecture)	Consistent

7. Business-IT Alignment

BUSINESS-IT ALIGNMENT	
BEFORE EA: • IT builds what they think • Business frustrated with IT • No clear roadmap • "IT doesn't understand us"	AFTER EA: • IT builds what business needs • Shared understanding • Architecture roadmap aligned • Stakeholder engagement

Real Example

Scenario	Before EA	After EA
Business request delivery	IT delivers something different - "This is not what we asked for!"	Architecture Vision document signed by stakeholders - Clear requirements, agreed scope, satisfied business

Summary Comparison Table

Area	BEFORE EA	AFTER EA
Technology	Everyone chooses their own	Standard catalog, approved choices
Integration	Spaghetti, point-to-point	Clean, documented, standard patterns
Decisions	Undocumented, repeated mistakes	ADRs, traceable, learn from history
Delivery Speed	Slow (from scratch) 6-8 weeks	Fast (reuse) 2-3 weeks
Cost	High (duplication), wasted resources	Lower (consolidation), right-sized
Security	Inconsistent, many vulnerabilities	Secure by design, standard controls
Business Alignment	"IT doesn't get us", misalignment	Shared vision, roadmap aligned

Key Metrics Improvement

KEY IMPROVEMENTS (Example Numbers)		
 Time to Market:	6-8 weeks → 2-3 weeks	(60% faster)
 Integration Time:	3-4 weeks → 2-3 days	(80% faster)
 Technology Variants:	7 databases → 2	(70% reduction)
 Applications:	150 → 85	(40% consolidated)
 Security Issues:	45 critical → 5	(90% reduction)
 Reuse Rate:	10% → 60%	(6x improvement)
 Stakeholder Satisfaction:	40% → 85%	(2x improvement)
 Cost Savings:	~\$2-3M per year	

Interview Tips

When asked about EA impact in interviews, structure your answer:

- 1. Pick 2-3 areas (e.g., Technology, Delivery Speed, Security)
- 2. Describe BEFORE state (the problems)
- 3. Describe AFTER state (the improvements)
- 4. Give specific metrics if possible
- 5. Share a real example from your experience

Sample Answer

"Before we implemented EA, each team chose their own technology stack. We had 7 different message brokers across the organization. After EA, we standardized on Kafka and SQS, reducing maintenance effort by 40% and saving significant licensing costs. Integration time for new services dropped from 3-4 weeks to 2-3 days because everyone now follows standard API patterns."

Conclusion

Enterprise Architecture transforms organizations from CHAOS to ORDER

Key Benefits:

- FASTER delivery (60%+ improvement)
- LOWER costs (consolidation, reuse)
- BETTER quality (standards, patterns)
- REDUCED risk (security, compliance)
- ALIGNED business and IT

EA is not overhead – it's an ENABLER for business success!